

CS 771A: Intro to Machine Learning, IIT Kanpur			Midsem Exam (22 Feb 2026)	
Name	MELBO			40 marks Page 1 of 4
Roll No	260007	Dept.	AWSM	

Instructions:

1. This question paper contains 2 pages (4 sides of paper). Please verify.
2. Write your name, roll number, department in **block letters** with **ink** on **each page**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – ambiguous cases will get 0 marks.



Q1 (Mending Melbo's Mistake) In Quiz 1, Melbo made a mistake while trying to simplify the SVM by maximizing the mean margin. The corrected optimization problem is given below. Assume that the data is such that perfect classifiers exist and $\sum_{i \in [N]} y^i \cdot \mathbf{x}^i \neq \mathbf{0}$. ((1+2) + 3 x 2 = 9 marks)

$$\min_{\mathbf{w}} \left\{ -\sum_{i \in [N]} y^i \cdot \mathbf{w}^T \mathbf{x}^i \right\} \quad \text{s.t.} \quad y^i \cdot (\mathbf{w}^T \mathbf{x}^i) \geq 0 \quad \forall i \in [n] \quad \text{and} \quad \|\mathbf{w}\|_2^2 \leq 1$$

Melba thinks the optimum solution $\mathbf{w}^*$ will always satisfy $\|\mathbf{w}^*\|_2 = 1$ even though the constraint asks for $\|\mathbf{w}\|_2^2 \leq 1$. Write **T** or **F**. If T, give proof. If F, give a counter example.

T

(Proof by contradiction). Suppose the optimum is $\mathbf{w}^*$ with $\|\mathbf{w}^*\|_2 = r < 1$. We claim $\tilde{\mathbf{w}} \stackrel{\text{def}}{=} \frac{\mathbf{w}^*}{r}$ is a better solution. Note that $y^i \cdot (\tilde{\mathbf{w}}^T \mathbf{x}^i) \geq 0$ for all $i \in [n]$ since $y^i \cdot (\mathbf{w}^{*T} \mathbf{x}^i) \geq 0$ for all $i \in [n]$. $\|\tilde{\mathbf{w}}\|_2 = 1$ i.e. $\tilde{\mathbf{w}}$ satisfies all constraints. But $\sum_{i \in [N]} y^i \cdot \tilde{\mathbf{w}}^T \mathbf{x}^i = \frac{1}{r} \cdot \sum_{i \in [N]} y^i \cdot \mathbf{w}^{*T} \mathbf{x}^i > \sum_{i \in [N]} y^i \cdot \mathbf{w}^{*T} \mathbf{x}^i$ since $y^i \cdot (\mathbf{w}^{*T} \mathbf{x}^i) \geq 0 \forall i$. Thus, $\mathbf{w}^*$ could not have been the optimum.

Write down the Lagrangian by introducing dual variables. No derivation needed.

$$\mathcal{L} = -\sum_{i \in [N]} (1 + \alpha_i) (y^i \cdot \mathbf{w}^T \mathbf{x}^i) + \beta (\|\mathbf{w}\|_2^2 - 1)$$

Give an expression for $\mathbf{w}$ in terms of the dual variables and constants e.g. $y^i, \mathbf{x}^i$ etc. No derivation needed.

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{0} \Rightarrow \mathbf{w} = \frac{1}{2\beta} \cdot \sum_{i \in [N]} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i$$

Simplify the dual problem (eliminate $\mathbf{w}$) – show major steps.

Eliminating $\mathbf{w}$ using its value computed above gives us

$$\mathcal{L} = -\frac{1}{4\beta} \cdot \left\| \sum_{i \in [N]} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i \right\|_2^2 - \beta$$

Thus, the dual problem is

$$\begin{aligned} \min & \frac{1}{4\beta} \cdot \left\| \sum_{i \in [N]} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i \right\|_2^2 + \beta \\ \text{s.t.} & \beta \geq 0 \quad \text{and} \quad \alpha_i \geq 0 \quad \forall i \in [n] \end{aligned}$$

Q2 (Multiclass Least Squares) Multiclassification can be solved as regression problems by assigning an embedding $\mathbf{v}^c \in \mathbb{R}^K$ to each class $c \in [C]$. Embeddings are collected in a matrix $V \in \mathbb{R}^{K \times C}$. For N train points $(\mathbf{x}^i, y^i), i = 1, \dots, N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in [C]$, the embedding assigned to the i^{th} point is $V\mathbf{e}_{y^i}$ where $\mathbf{e}_c \in \mathbb{R}^C$ is the standard basis vector with 1 at the c^{th} coordinate and 0 elsewhere. Melba wants to learn K models $\mathbf{w}^k \in \mathbb{R}^d$, collected as a $K \times d$ matrix $W \in \mathbb{R}^{K \times d}$ by solving $\min_{W, V} \frac{1}{2} \sum_{k \in [K]} \|\mathbf{w}^k\|_2^2 + \frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2 + \sum_{i \in [N]} \|W\mathbf{x}^i - V\mathbf{e}_{y^i}\|_2^2$ using alternating optimization.

a. Keeping the embeddings V fixed, find the optimal value of $W \in \mathbb{R}^{K \times d}$. Show steps. (4 marks)

Rewriting using Frobenius norm notation gives us $\min_W \frac{1}{2} \|W\|_F^2 + \sum_{i \in [N]} \|W\mathbf{x}^i - V\mathbf{e}_{y^i}\|_2^2$. Setting the derivative to zero gives us $W + 2 \cdot \sum_{i \in [N]} (W\mathbf{x}^i - V\mathbf{e}_{y^i})(\mathbf{x}^i)^\top = \mathbf{0}$ i.e.

$$W = 2 \cdot \left(I + 2 \sum_{i \in [N]} \mathbf{x}^i (\mathbf{x}^i)^\top \right)^{-1} \sum_{i \in [N]} V\mathbf{e}_{y^i} (\mathbf{x}^i)^\top$$

b. Keeping the models W fixed, find the optimal value of $V \in \mathbb{R}^{K \times C}$. Show steps. (4 marks)

Similar to part a. Rewrite using Frobenius norm notation $\min_V \frac{1}{2} \|V\|_F^2 + \sum_{i \in [N]} \|W\mathbf{x}^i - V\mathbf{e}_{y^i}\|_2^2$. Set the derivative to zero to get $V + 2 \cdot \sum_{i \in [N]} (V\mathbf{e}_{y^i} - W\mathbf{x}^i)(\mathbf{e}_{y^i})^\top = \mathbf{0}$ i.e.

$$V = 2 \cdot \left(I + 2 \sum_{i \in [N]} \mathbf{e}_{y^i} (\mathbf{e}_{y^i})^\top \right)^{-1} \sum_{i \in [N]} W\mathbf{x}^i (\mathbf{e}_{y^i})^\top$$

For part c. the first option would actually encourage embeddings to get closer, the second option would encourage them to be farther, the third option is just the second option scaled down, the fourth option minimizes $\sum_{c,d} \|\mathbf{v}^c + \mathbf{v}^d\|_2^2$ encouraging embeddings to collapse.

c. Melbo notes that Melba's objective doesn't encourage the embeddings $\mathbf{v}^c, c \in [C]$ to be far apart from each other. Melbo wants to use a different regularizer by replacing the term $\frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2$. Which of the following regularizers (if added to Melba's objective after removing $\frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2$) will encourage the embeddings to be far apart? Fill/shade one or more circles (no derivation needed). Note: $\mathbf{1} \in \mathbb{R}^C$ is the all-ones vector. (4 marks)

$$+ \frac{1}{2C} \sum_{c \in [C]} \sum_{d \in [C]} \|\mathbf{v}^c - \mathbf{v}^d\|_2^2$$



$$- \frac{1}{2C} \sum_{c \in [C]} \sum_{d \in [C]} \|\mathbf{v}^c - \mathbf{v}^d\|_2^2$$



$$- \frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2 + \frac{1}{2C} \cdot \|V\mathbf{1}\|_2^2$$



$$+ \frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2 + \frac{1}{2C} \cdot \|V\mathbf{1}\|_2^2$$



None of the above



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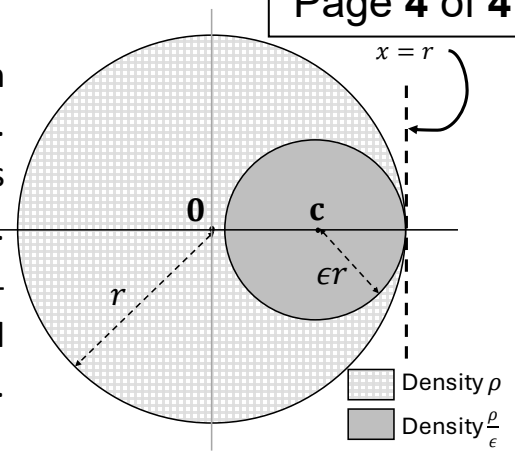
Q3. (True-False) Write **T** or **F** for True/False in the box on the right and a brief justification below.

Note: The expression $p \gg q$ means p is “much larger” than q .

(3 x (1+2) = 9 marks)

1	Using the tanh function $\tau(c) \stackrel{\text{def}}{=} \frac{e^{2c}-1}{e^{2c}+1}$, defining $\mathbb{P}[y^i \mathbf{x}^i, \mathbf{w}] \stackrel{\text{def}}{=} \tau(\mathbf{y}^i \cdot \mathbf{w}^\top \mathbf{x}^i)$ is a valid choice of likelihood for a binary classification problem with $\{-1, +1\}$ labels.	F
Probability distributions must be non-negative and be normalized (sum/integrate to unity). The tanh function takes negative values $\tau(c) < 0$ for $c < 0$. The tanh function is anti-symmetric i.e. $\tau(-c) = -\tau(c)$ which means that $\mathbb{P}[1 \mathbf{x}^i, \mathbf{w}] + \mathbb{P}[-1 \mathbf{x}^i, \mathbf{w}] = \tau(\mathbf{w}^\top \mathbf{x}^i) + \tau(-\mathbf{w}^\top \mathbf{x}^i) = 0$ whereas we want $\mathbb{P}[1 \mathbf{x}^i, \mathbf{w}] + \mathbb{P}[-1 \mathbf{x}^i, \mathbf{w}] = 1$. Thus, the tanh function is not a valid choice.		
2	For a data point $(\mathbf{x}, y)$, redefining softmax loss using the square function looks like $-\ln\left(\frac{\eta_y}{\sum_{c \in [C]} \eta_c}\right)$ with $\eta_c \stackrel{\text{def}}{=} (\mathbf{w}_c^\top \mathbf{x})^2$. If this loss goes to zero for $(\mathbf{x}, y)$, then it must be the case that $\mathbf{w}_y^\top \mathbf{x} \gg \mathbf{w}_c^\top \mathbf{x}$ for all $c \neq y$. Give a proof if T or a counterexample if F .	F
Consider a toy problem with 2D features and three classes r, g, b for which the classifiers are $\mathbf{w}_r = (-10, -10), \mathbf{w}_g = (-10, +10), \mathbf{w}_b = (+10, -10)$ Consider a data point with $\mathbf{x} = (1, 1), y = r$ i.e., $\eta_r = 400, \eta_g = 0, \eta_b = 0$. The softmax loss is $-\ln\left(\frac{\eta_r}{\eta_r + \eta_g + \eta_b}\right) = -\ln\left(\frac{400}{400}\right) = 0$ but clearly $\mathbf{w}_r^\top \mathbf{x} = -20 \ll \mathbf{w}_g^\top \mathbf{x} = \mathbf{w}_b^\top \mathbf{x}$. Thus, this version of softmax loss may encourage weird behavior in the learnt classifier. To avoid this weirdness, monotonic functions are usually used e.g. $\exp(\)$, $\text{ReLU}(\)$.		
3	A bin with N red, N green and N blue balls has entropy $\log_2 3$. Moreover, no matter how many red/green/blue balls are added to the bin hereon (no other colour is allowed, no ball is taken out), the entropy of the bin can never fall below $\log_2 3$.	F
The initial proportions of the bin are $\left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$ for which the entropy is indeed $\log_2 3$. Suppose we add $\exp(100N)$ (or some similarly stupidly large number of) red balls (no green or blue balls are added). The proportions of the bin now approach $[1, 0, 0]$ i.e. the entropy approaches 0.		

Q4 (Insert Clever Name) Melbruh has a 2D distribution $\mathcal{D}$ with support inside a unit area circle of radius r centered at the origin. $\mathcal{D}$ is described by parameters $\rho \geq 0$ and $\epsilon \in [0,1]$. The density is $\frac{\rho}{\epsilon}$ in the *dense* circle of radius ϵr whose center $\mathbf{c}$ is on the x-axis. The density is ρ in the rest of the support. The two circles are co-tangential with the line $x = r$ as common tangent. Find out valid range of ρ for which $\mathcal{D}$ is a proper dist. with value of $\epsilon \in [0,1]$. Find an expression for $\epsilon \in [0,1]$ in terms of ρ and constants.



For valid combinations of ρ, ϵ , find mean vector $\boldsymbol{\mu} \in \mathbb{R}^2$ of $\mathcal{D}$. *Hint: The mean of the uniform dist. over a circle is its centre.* Give final answers in boxes, derivations later. **(2 + 2 + 3 + 3 = 10 marks)**

$$\boxed{4/5} \leq \rho \leq \boxed{1}$$

$$\epsilon = \left(1 \pm \sqrt{1 + 4\left(1 - \frac{1}{\rho}\right)}\right)/2$$

$$\boldsymbol{\mu} = (\epsilon - \epsilon^2) \cdot \rho \cdot \mathbf{c} \equiv (1 - \rho) \cdot \mathbf{c}$$

Give brief derivation for valid ranges of ρ , valid values of ϵ for which $\mathcal{D}$ is a proper distribution and its mean $\boldsymbol{\mu}$.

Note that $\pi r^2 = 1$. Normalization dictates $\pi r^2(1 - \epsilon^2) \cdot \rho + \pi r^2 \epsilon^2 \cdot \frac{\rho}{\epsilon} = 1$. This means $\epsilon = \left(1 \pm \sqrt{1 + 4\left(1 - \frac{1}{\rho}\right)}\right)/2$. For ϵ to be real, we need $1 + 4\left(1 - \frac{1}{\rho}\right) \geq 0$ which gives $\rho \geq \frac{4}{5}$. For ϵ to be between 0 and 1, we need $1 + 4\left(1 - \frac{1}{\rho}\right) \leq 1$ which gives $\rho \leq 1$. Let $\mathcal{S}$ be the small circle with density $\mathcal{D}(\mathbf{x}) = \frac{\rho}{\epsilon}$ and $\mathcal{C}$ be large circle with density $\mathcal{D}(\mathbf{x}) = \rho$ in $\mathcal{C} \setminus \mathcal{S}$. We have $\boldsymbol{\mu} = \int_{\mathcal{C}} \mathbf{x} \cdot \mathcal{D}(\mathbf{x}) d\mathbf{x} = \underbrace{\int_{\mathcal{S}} \mathbf{x} \cdot \mathcal{D}(\mathbf{x}) d\mathbf{x}}_{(A)} + \underbrace{\int_{\mathcal{C} \setminus \mathcal{S}} \mathbf{x} \cdot \mathcal{D}(\mathbf{x}) d\mathbf{x}}_{(B)}$. Now, $(A) = \frac{\rho}{\epsilon} \int_{\mathcal{S}} \mathbf{x} d\mathbf{x}$ and $\int_{\mathcal{S}} \mathbf{x} d\mathbf{x} = \pi r^2 \epsilon^2 \cdot \int_{\mathcal{S}} \mathbf{x} \cdot \mathcal{U}(\mathbf{x}) d\mathbf{x}$ where $\mathcal{U}(\mathbf{x}) = \frac{1}{\pi r^2 \epsilon^2}$ is the (conditional) uniform dist. on $\mathcal{S}$. Using the

hint gives us $(A) = \frac{\rho}{\epsilon} \cdot \pi r^2 \epsilon^2 \cdot \mathbf{c} = \rho \cdot \epsilon \cdot \mathbf{c}$. Also, $(B) = \rho \int_{\mathcal{C} \setminus \mathcal{S}} \mathbf{x} d\mathbf{x} = \rho \left(\underbrace{\int_{\mathcal{C}} \mathbf{x} d\mathbf{x}}_{(C)} - \underbrace{\int_{\mathcal{S}} \mathbf{x} d\mathbf{x}}_{(D)} \right)$.

Using the same argument as above, we get $(C) = \pi r^2 \cdot \mathbf{0}$ and $(D) = \pi r^2 \epsilon^2 \cdot \mathbf{c}$ which gives us $(B) = -\rho \cdot \pi r^2 \epsilon^2 \cdot \mathbf{c} = -\rho \cdot \epsilon^2 \cdot \mathbf{c}$ giving $\boldsymbol{\mu} = (\epsilon - \epsilon^2) \cdot \rho \cdot \mathbf{c}$. This can be simplified by reusing the expression for ϵ in terms of ρ derived earlier $(\epsilon - \epsilon^2) \cdot \rho = 1 - \rho$, giving $\boldsymbol{\mu} = (1 - \rho) \cdot \mathbf{c}$.

This can be further simplified by using $\mathbf{c} = \left(\frac{1}{\sqrt{\pi}}(1 - \epsilon), 0\right)$ as the circles are co-tangential giving

$\boldsymbol{\mu} = \left(\frac{1}{\sqrt{\pi}}(1 - \rho)(1 - \epsilon), 0\right) = \left(\frac{1}{2\sqrt{\pi}}(1 - \rho) \left(1 \mp \sqrt{1 + 4\left(1 - \frac{1}{\rho}\right)}\right), 0\right)$. Either answer is okay.